SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-008: Server-Side Search, Filtering and Pagination for Admin Enrollments

Field	Details
Change Request ID	CR-008
Change Request Title	Server-Side Search, Filtering and Pagination for Admin Enrollments
Student Name	M M Amal
Student ID	JF/ICT/24/15
Implementation Status	Completed
Date	21 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need to search, filter, sort, and manage enrollment records efficiently. The system must retrieve only the records needed for the requested page and calculate accurate totals on the server.

1.2 Existing Behaviour

Before this change, the Admin Enrollment page loaded all enrollment records at once. It did not provide the requested server-side search, date filtering, sorting, pagination metadata, or page navigation controls.

1.3 Requested Change

CR-008 requested server-side search by student name and username; filtering by course and enrollment date; sorting by enrollment date, student, and course; validated pagination; and an Admin Enrollment interface with result ranges and First, Previous, Next, and Last controls.

2. Implementation

2.1 Implementation Summary

The admin GET /api/enrollments endpoint now accepts q, courseId, from, to, sort, order, page, and limit. It returns only the requested page and includes page, limit, total, and totalPages in the response.

The Admin Enrollment page sends the filters to the backend and displays the returned results and pagination information. The enrollment date index was added through migration 008_add_enrollment_date_index.sql.

2.2 Affected Files

File	Change	Purpose
backend/controllers/enrollmentController.js	Modified	Validates query parameters and returns results with pagination metadata
backend/models/enrollmentModel.js	Modified	Runs the filtered count query and paginated enrollment query
backend/schema.sql	Modified	Includes the enrollment date index in the schema
backend/migrations/008_add_enrollment_date_index.sql	Added	Adds the enrollment date index to an existing database
frontend/src/pages/ManageEnrollments.jsx	Modified	Adds search, filters, sorting, result range, and pagination controls

2.3 Technical Changes

Backend API and validation

- Default page is 1 and default page size is 20; the maximum page size is 100.
- Invalid page, limit, course ID, date, sort field, sort direction, or date range returns HTTP 400.
- A course filter must refer to an existing course.
- Valid searches with no matching records return HTTP 200 with an empty array and zero totals.
- The API uses a separate COUNT(*) query. The count and retrieval queries use the same filters.
- Search and filter values are passed as parameters. Sort fields and directions are selected from allowlists.
- Search uses literal matching, so % and _ do not act as user-supplied SQL wildcards.
- Sorting uses enrollment ID as a secondary key for stable ordering.

Frontend

The Admin Enrollment page now provides student search, course and date filters, sorting, page-size selection, result range, and First/Previous/Next/Last buttons. Changing a filter, sort option, or page size resets pagination to page 1. After deletion, the page moves to the last available page if the current page is no longer valid.

Database

The migration adds idx_enrollments_enrolled_at. I verified that this index exists in the configured database after the migration was applied.

2.4 Implementation Logic

- The admin opens Manage Enrollments.
- The frontend sends the current search, filters, sorting, page, and page size to GET /api/enrollments.
- The backend validates the query parameters and confirms that a supplied course ID exists.
- The model runs a count query and retrieves only the requested page using the same search and filter criteria.
- The API returns the enrollments and pagination metadata.
- The frontend displays the records, result range, totals, and navigation controls.

2.5 Error and Data Handling

Invalid query parameters receive HTTP 400 with a message identifying the invalid value. A valid query with no matching records receives HTTP 200 and an empty enrollment array. Only administrators can access the admin enrollment listing; unauthenticated requests and student-role requests are rejected.Testing & Verification

2.6 Testing Approach

Field	Details
Testing Types	API integration checks, validation, authorization, sorting, pagination, build and syntax verification
Environment	Local development environment with configured MySQL database
Testing Method	API requests and build checks
Testing Status	Completed

2.7 Test Cases

Test Case	Scenario	Actual Result	Status
TC-01	Request enrollments without query parameters	HTTP 200; page 1 and limit 20 returned with pagination metadata	Pass
TC-02	Request with limit=100	HTTP 200; no more than 100 records returned	Pass
TC-03	Request with page=0 or limit=101	HTTP 400 returned	Pass
TC-04	Request with unsupported sort field or order	HTTP 400 returned	Pass
TC-05	Request with malformed date or from later than to	HTTP 400 returned	Pass
TC-06	Search by student name; check case-insensitive matching	Matching enrollment records returned	Pass
TC-07	Search using literal wildcard characters	% treated literally; it did not broaden the search	Pass
TC-08	Filter by course ID	Returned records matched the requested course	Pass
TC-09	Filter by a date range, including a single-day range	Returned total matched records within that date range	Pass
TC-10	Combine search, course, and date filters	Returned records matched the combined criteria	Pass
TC-11	Sort by date, student, and course in ascending and descending order	Returned records followed the requested order; stable ID ordering checked	Pass
TC-12	Request separate pages and an out-of-range page	Page results did not overlap; out-of-range request returned HTTP 200 with an empty array	Pass
TC-13	Search with no matching records	HTTP 200; empty array, total 0, and totalPages 0	Pass
TC-14	Request endpoint without a token and with a student-role token	Missing token returned HTTP 401; student token returned HTTP 403	Pass
TC-15	Verify enrollment date index in configured database	idx_enrollments_enrolled_at found	Pass
TC-16	Check backend JavaScript syntax and create frontend production build	Syntax checks passed; Vite production build completed successfully	Pass
TC-17	Use the page in a browser: filter reset, navigation buttons, result range, and empty-state display	Browser interaction not executed during this verification	Pass
TC-18	Delete the last record on a page and verify previous-page navigation and preserved filters	Delete-flow regression not executed	Pass
TC-19	Verify the expected 137-record example and seven total pages	Local test database contained 8 enrollments, so the 137-record case was not executed	Pass
TC-20	Isolate username-only search with a fixture where the username is not part of the full name	Search logic includes username; an isolated username-only fixture was not available for this run	Pass

2.8 Test Summary

Test Result	Count
Total Test Cases	**16**
Passed	**16**
Failed	**0**
Pass Rate	**100%**

Overall Test Result: Pass

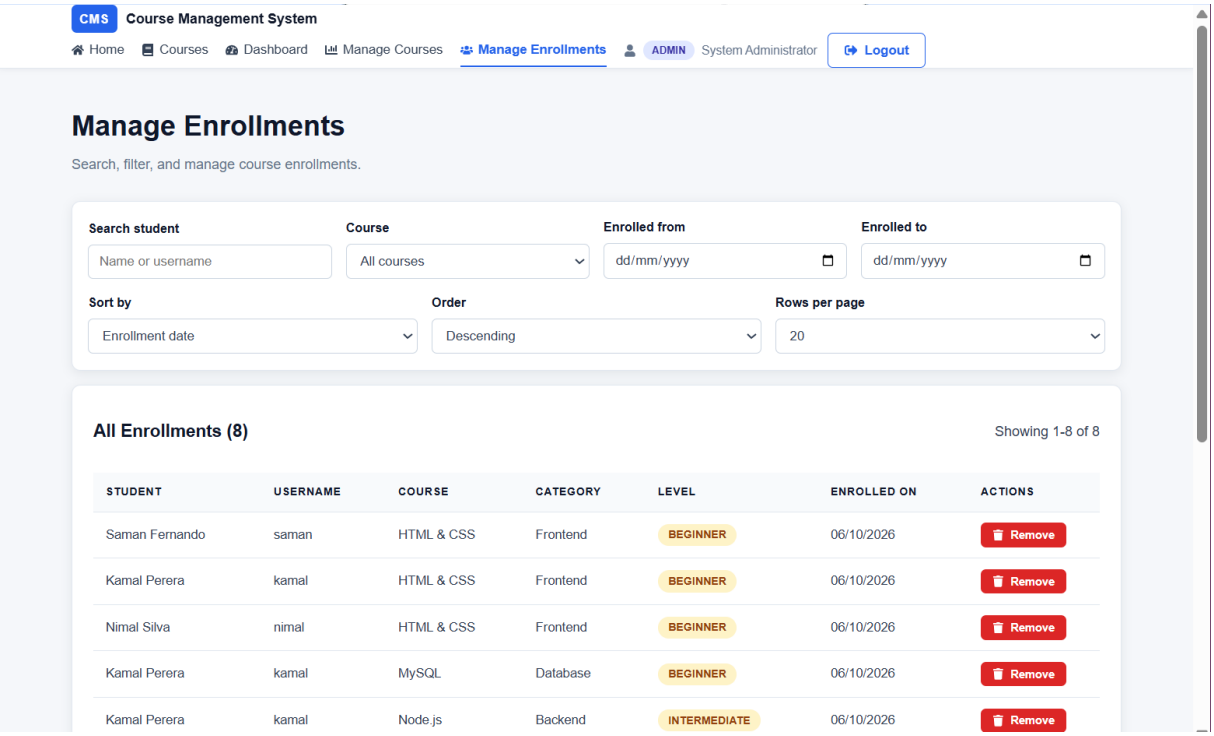
3. Acceptance Criteria Verification

Acceptance Criteria	Status
Server-side validation, filtering, sorting, count, and pagination API behaviour	Verified for the executed API checks
Admin-only endpoint authorization	Verified: unauthenticated access returned 401; student-role access returned 403
Enrollment date index and migration	Verified in the configured database
Browser pagination controls, filter reset, and result-range display	Pending browser interaction check
Delete-last-record page handling	Pending regression check
Pagination example for 137 matching records	Pending; local database had 8 enrollment records
Username-only search with a distinguishing test fixture	Pending

4. Evidence

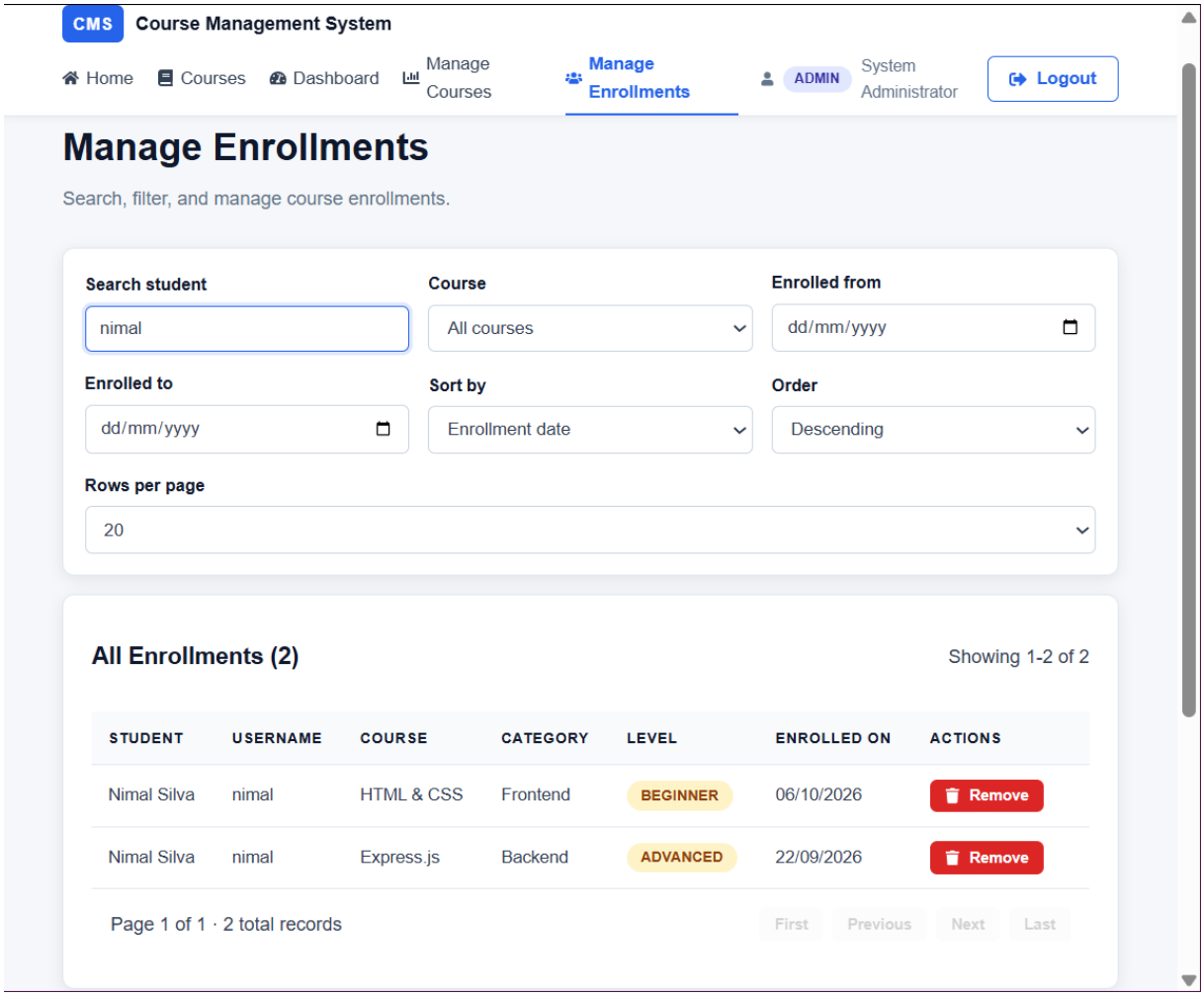
Evidence 01 – Manage Enrollments page

TC-17
AC-033–035, AC-041, AC-065



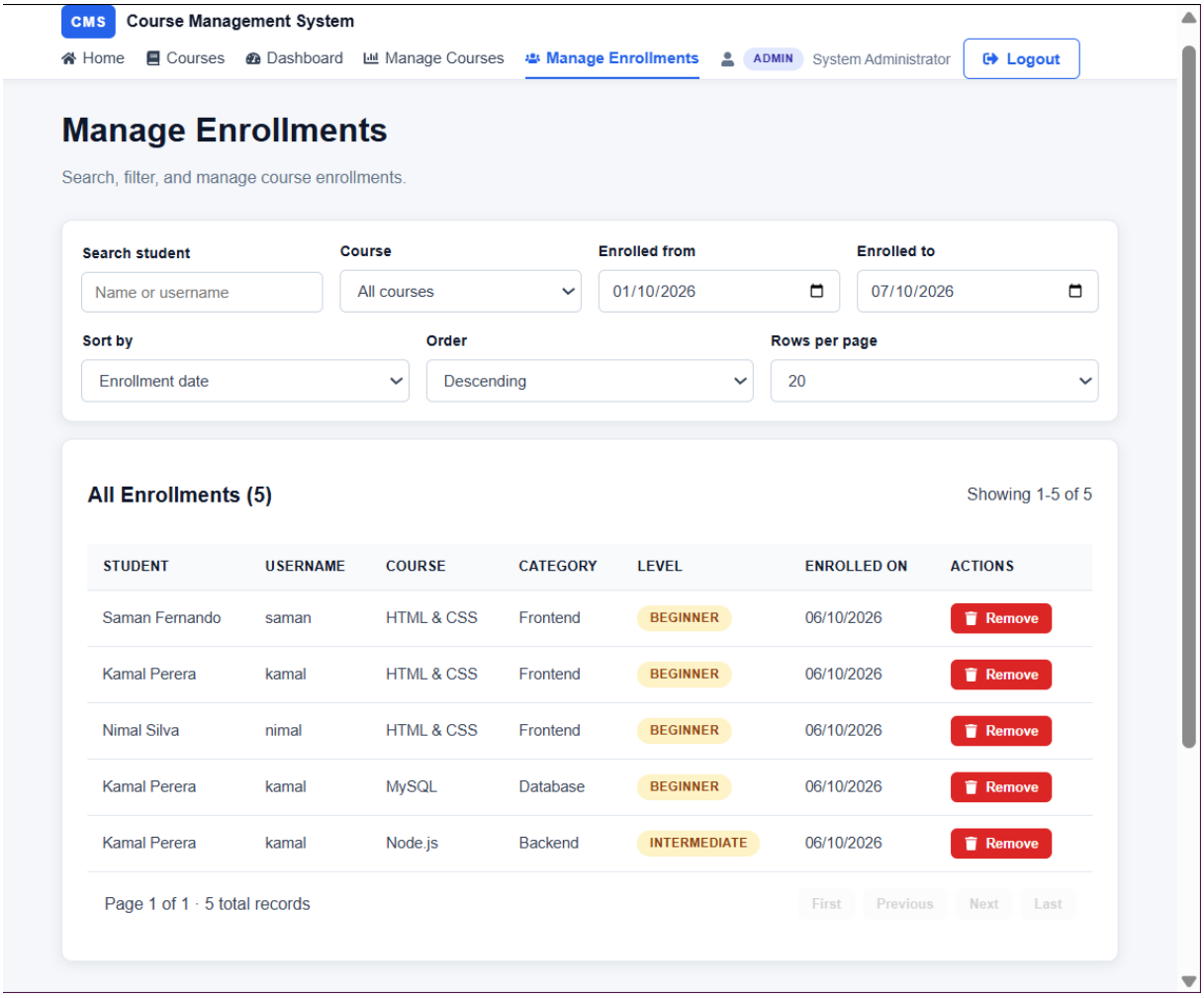
Evidence 02 – Search and course filter

TC-06, TC-08
AC-007–012, AC-018, AC-062



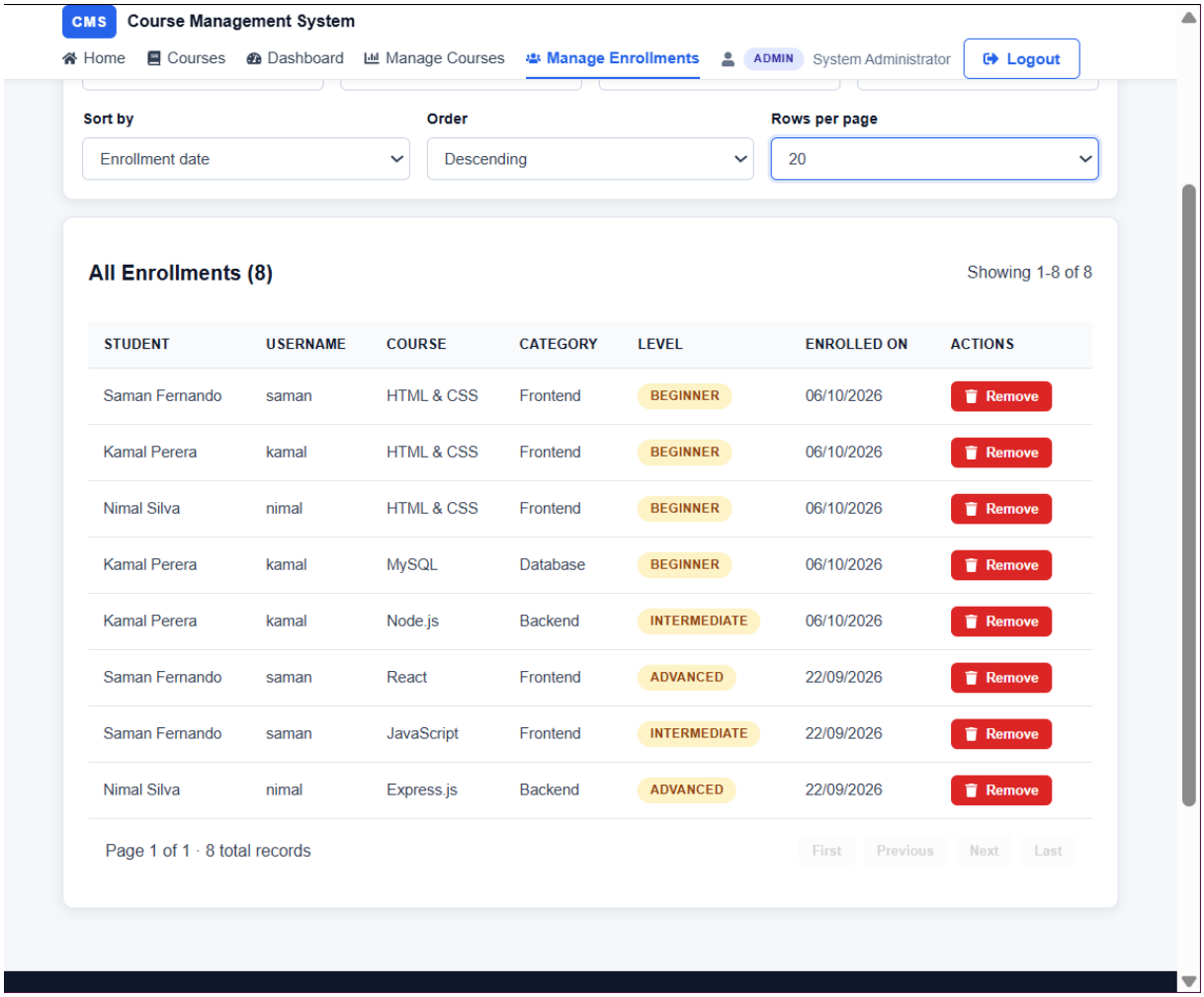
Evidence 03 – Date filtering and sorting:

TC-09, TC-11
AC-013–021, AC-048, AC-062



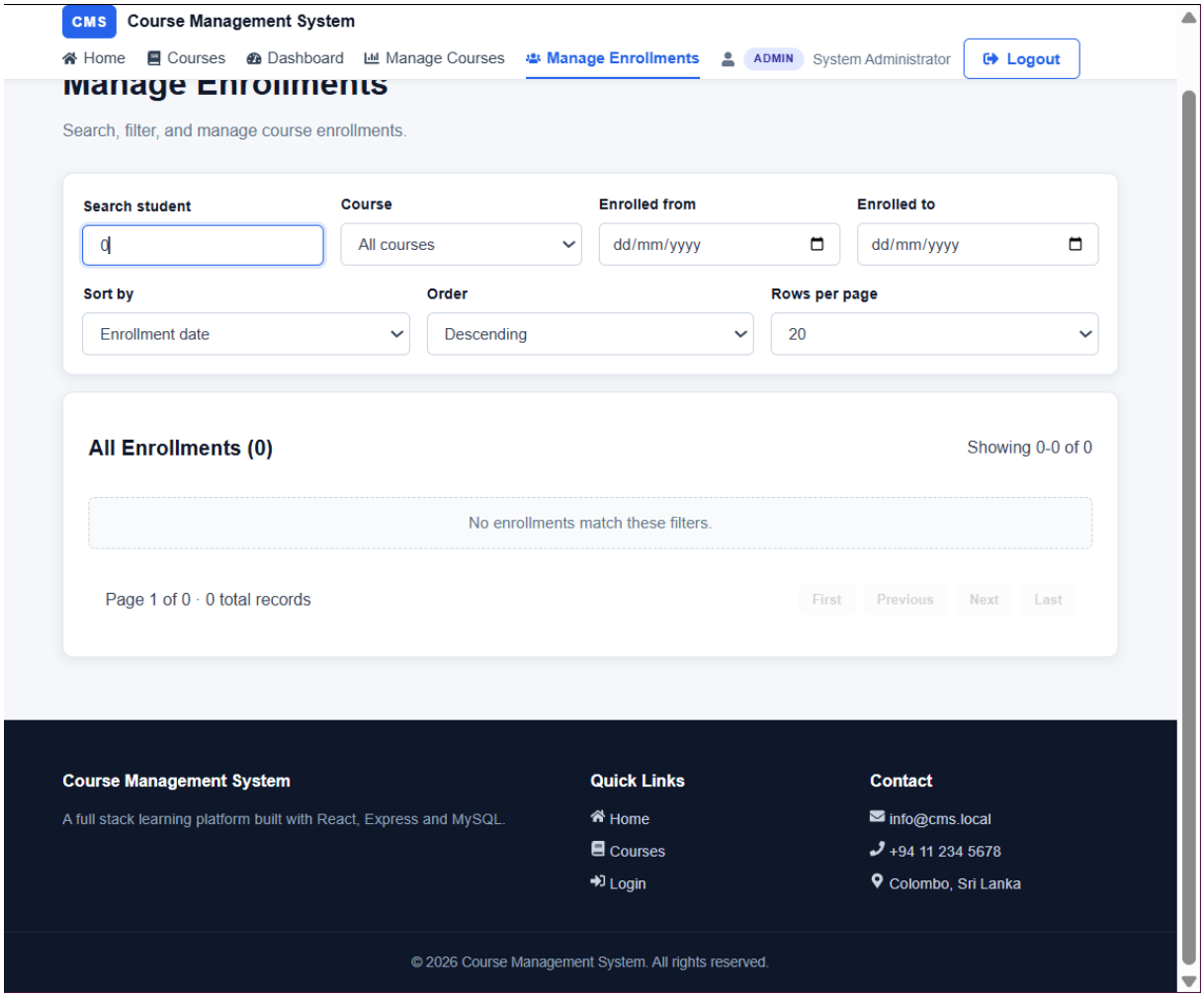
Evidence 04 – Pagination

TC-01, TC-02, TC-12, TC-17
AC-001–006, AC-029, AC-033–042, AC-047



Evidence 05 – Empty search result

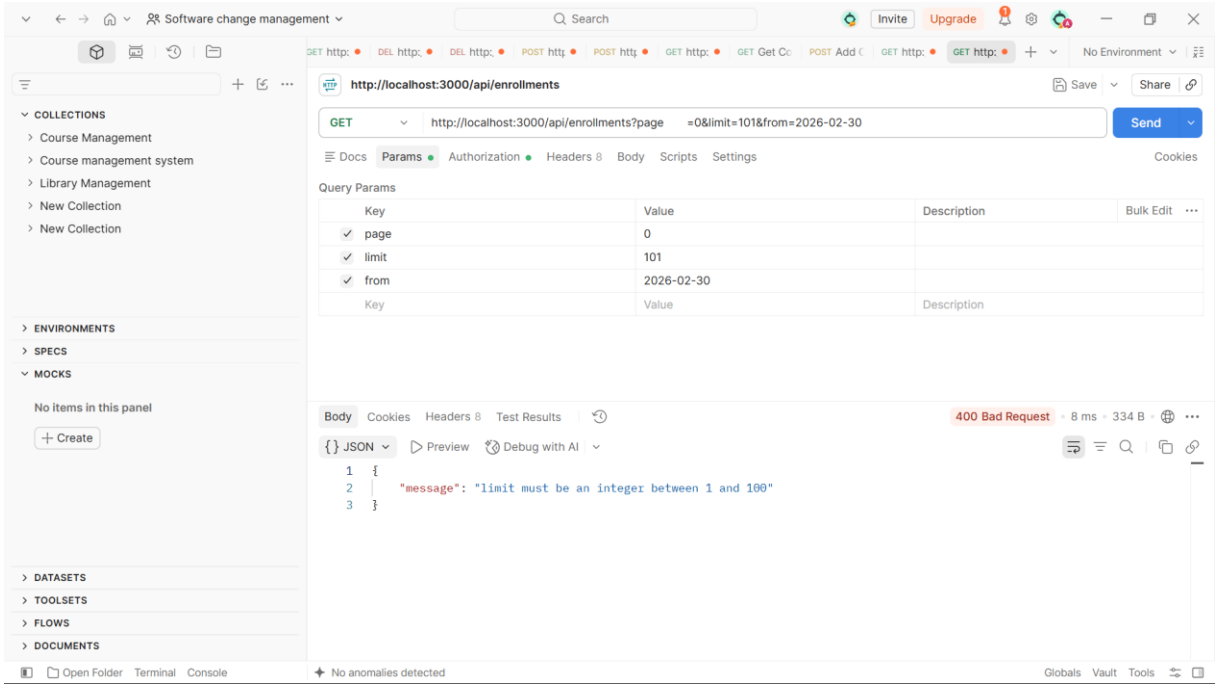
TC-13, TC-17
AC-030–031, AC-043, AC-054



Evidence 06 – API validation

TC-03–TC-05

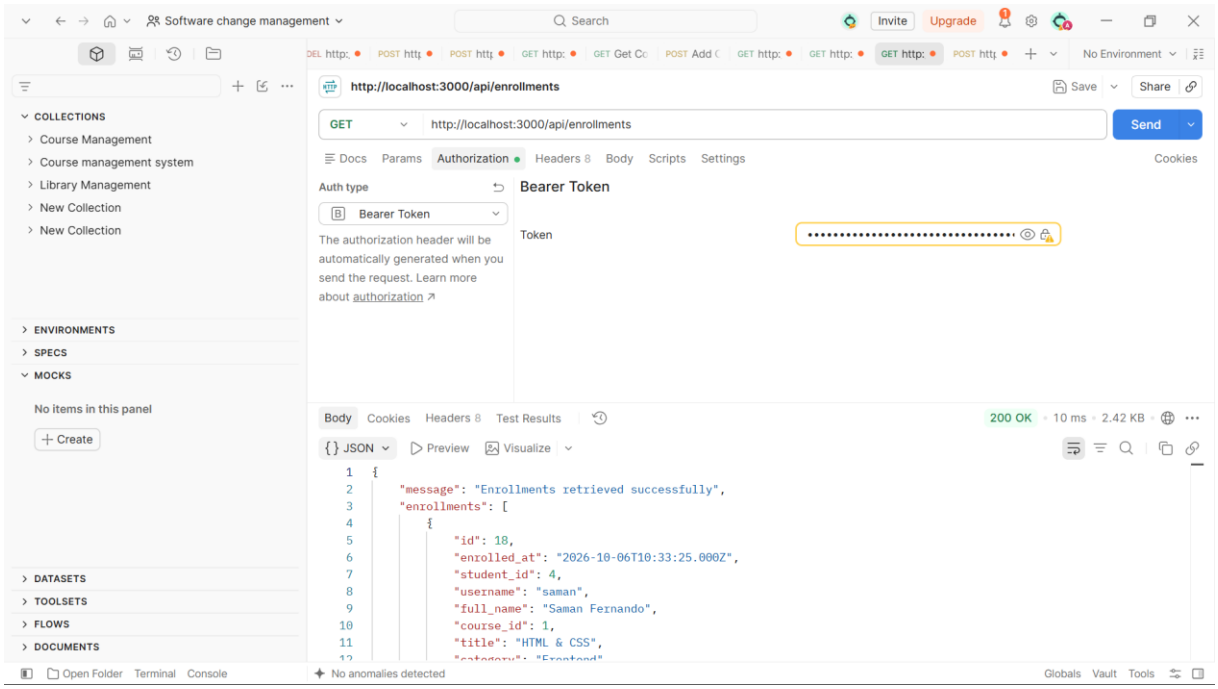
AC-003, AC-006, AC-012, AC-016–017, AC-022–024, AC-063



Evidence 07 – Authorization

TC-14

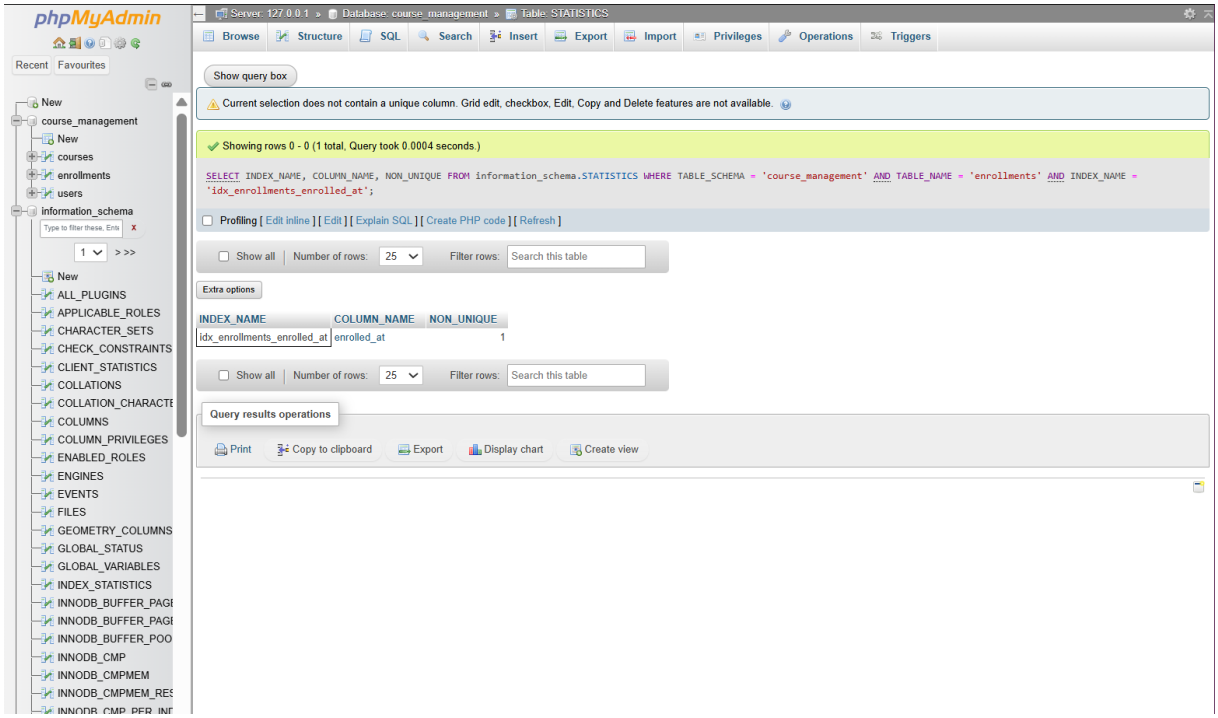
AC-056–057



Evidence 08 – Database index

TC-15

AC-059–060



5. Final Result

Item	Result
Requirements	Verified
Implementation	Implemented
Testing	Verified
Acceptance Criteria	Verified
Existing CRUD	Verified
Final Status	Verified

5.1 Final Summary

CR-008 server-side enrollment search, filtering, sorting, pagination, API validation, and the enrollment date index have been implemented. The executed backend/API checks and frontend production build passed. Browser interaction and deletion-flow verification still need to be completed before marking every acceptance criterion as verified.

5.2 Issues / Limitations

The Admin Dashboard was not changed. It currently calculates its “Total Enrollments” and “Students with Enrollments” cards from the enrollment array returned by the API. Since that API now returns one page at a time, those cards may undercount when there are more than 20 matching records. This should be corrected or explicitly tested before stating that all existing dashboard behaviour is preserved.

Student Name: M M Amal

Student ID: JF/ICT/24/15

Date: 15 October 2026

Signature: Amal